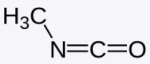
Introduction to Computational Chemistry & Biochemistry
Homework 1: Molecular Representations

Recommend using Jupyter (Colab) Notebook with numpy, RDKit, py3Dmol (and PubChem database).

1. Report the SMILES strings of (a) molecular hydrogen, (b) water, (c) heavy water, (d) carbon dioxide, (e) methane, (f) butane, (g) acetic acid, (h) cyclohexane, (i) benzene, (j) pyridine, (k) pyrimidine, (l) benzoic acid, (m) salicylic acid, (n) aspirin, (o) D-proline, (p) zwitterionic tyrosine, (q) o-phosphotyrosine, (r) sucrose , (s) vitamin B_1, (t) cephalostatin I, (u) ciclosporin, (v) human insulin.
2. Render 2D MOL molecule representations of (a)-(u) and convert to svg and png format, which are most suitable for creating a table of molecular name, SMILES string, 2D molecular structure. As an example, here's a screenshot from wikipedia:

Molecule	Structure	SMILES formula
Dinitrogen	N≡N	`N#N`
Methyl isocyanate (MIC)		`CN=C=O`
Copper(II) sulfate	$Cu^{2+}SO_4^{2-}$	`[Cu+2].[O-]S(=O)(=O)[O-]`
Vanillin		`O=Cc1ccc(O)c(OC)c1` `COc1cc(C=O)ccc1O`

3. Creating clear informative 2D molecular structures often requires choosing whether or not to implicitly or explicitly show hydrogens as well as labels, coloring, bonding, etc. For example, SMILES strings typically omit hydrogen atoms to keep the text short. Add the implicit hydrogens to (e)-(o) and render 2D MOL molecular structures. Make a table with the 2D molecular structure images (svg, png) of (e)-(o) showing a comparison of implicit and explicit renderings.
4. Render 3D MOL molecular structures of (a)-(u) from the SMILES strings using open-source cheminformatics or molecular visualization tools (e.g, RDKit, Open Babel, Avogadro, MolView). SMILES strings do not contain 3D spatial coordinates by default. To export a 3D structure, you need to embed hydrogens and generate a 3D conformer (typically with ETKDG and/or molecular mechanics optimized geometry using MMFF94 force field). Either generate a zip file with the .mol and .pdb files, or *preferably*, show the 3D structures in a jupyter (google colab) notebook.
5. Representing nucleotide (DNA/RNA) or amino acid (protein, peptide) sequences more commonly uses FASTA strings (not SMILES). Report the FASTA strings of (v). Also, the 3D viewing of large peptides and proteins commonly uses implicit backbone ribbon views instead of explicitly showing all atoms and associated bonding. Show a common 3D 'ribbon' structural view of (v) insulin.